AI Application R&D Engineer · BenQ Materials

Shan-Cheng Chang 張善程

LLM Agent Engineering · AI Data Analytics · Document AI

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— CORE SKILLS

Agent Engineering (live)
Skills (Claude Code style) · Sandboxing · RBAC / Audit Log

Agent Prototypes (PoC)
Agent Harness · Function Calling · MCP · Context Management · Knowledge Graph (PostgreSQL + pgvector)

Evaluation & Validation
Golden Set · Code Validators

LLM
Azure OpenAI · Gemini API · Prompt Engineering · APO

Document AI / ML / CV
Azure Document Intelligence · OCR + LLM Extraction · YOLOv8 · PyTorch

Backend & Web
Python · FastAPI · Node.js · React · JavaScript / TypeScript · PostgreSQL · SQL Server

DevOps & Tools
Docker · GCP · Azure · Git · n8n · Claude Code

Security
AES-256-GCM · JWT · AD / Entra ID · ISO 27001-aligned

— EDUCATION

National Taiwan University
M.S. Biomechatronics Eng. (AIoT / CV) · 2021 – 2023

Taipei Tech (NTUT)
B.S. Mechanical Engineering · 2017 – 2021

AI application engineer who designs, validates and maintains enterprise AI systems: architect of BenQ Materials' AI data-analysis platform (in daily production use) and its document-AI stack, built from zero. Every system is checked against a test set or the system it replaces before go-live.

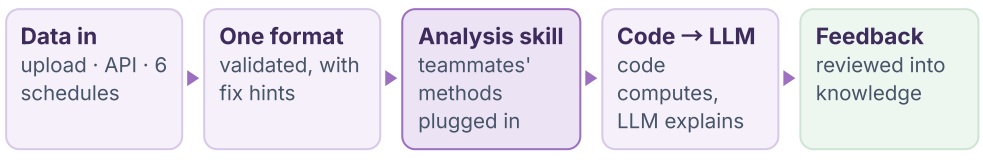
4.15M NTD	6	~1M NTD	1,500+ h
AUTOML AVOIDED / 2 YRS	LINE FLOWS ON PLATFORM	BENEFIT / YR (3 OCR)	SAVED / YR (3 OCR)

— EXPERIENCE

AI Application R&D Engineer — Dec 2024 – Present

BenQ Materials Corp. (明基材料) · Digital Technology Center, Intelligent Application Dev. Sec. · Taoyuan

- **Enterprise Brain → AI data-analysis platform** (system architect & lead developer) — used daily by the BML line, replacing a purchased AutoML package (**4.15M NTD / 2 yrs**); 6 flows cut over with zero downtime. FastAPI + Azure OpenAI (company-hosted API) + Docker sandbox.



Data → analysis → knowledge. Purple box: the shared skill I designed.

	Before (n8n)	After (one skill)
Analysis logic	6 flows, ~550 nodes	**1 skill, 3 entry points**
New method	edit every flow	**teammate's logic, wired once**
Migration check	—	**Old = new: Excel 9/9 cells; same PDFs**

  - **Why a skill** — hard-coding would work today; a Claude Code–format skill lets a future chat LLM call the same methods. Analysis already runs in a Docker sandbox that chat will reuse.
  - **Status** — analysis platform live; now building an OCAP text knowledge base (K1, incl. my 6 flows) and a user-feedback base (K2) before chat; MCP and knowledge graph are prototypes. Full architecture on my website.
- **Document-AI applications (OCR + LLM)** — built the company's platform from zero (Azure Document Intelligence + Azure OpenAI); every system scored field-by-field in code on a hand-labelled test set before go-live.

System	Accuracy (test set)	Impact
Quotation OCR	validity **51% → 100%** (29 docs)	30 → 10 s; 758+ docs
Invoice OCR (TW/CN)	**31/32**, **31/31** fully correct	1,447 h/yr; CN: 3 sites
Health-check OCR	**97.4%** fields (42 × 46)	AES-256, AD; ISO 27001-aligned
APO-OCR prompt tuning	validating (32-invoice set)	team tool, pre-launch

— LANGUAGES

Mandarin Chinese — Native
English — Professional (TOEIC 790)
German — A2

— CERTIFICATIONS

Company Digital Talent L3 (Expert)
Military service completed

— EARLIER PROJECT

Autonomous navigation with Autoware / ROS
B.S. capstone, NTUT · 2020 – 2021
Indoor/outdoor self-driving platform (mapping, localization, path planning) with an ML mask-detection module.

— PUBLICATIONS & PATENT

Chou, C.-Y., **Chang, S.-C.**, et al. (2023). Development of AIoT system for facility asparagus cultivation. *Computers and Electronics in Agriculture*, 206:107665. *(co-first author with advisor)*

Chang, S.-C., et al. (2022). AIoT based pest counting system for asparagus cultivation. *Proc. 10th ISMAB*.

Taiwan Invention Patent TW I842416 B — Crop Harvester, 2024.

— EXPERIENCE (CONT.)

BenQ Materials — other systems — Dec 2024 – Present

- **Google Cloud Platform — supplier questionnaire management system** (live) — 52 APIs, built solo in 4.5 months, handed over; React 19 + Cloud Functions + Entra ID.
- **ESG carbon platform refactor & handover** — import moved to a web upload run by the user department; backend split into one module per page (APIs 104 → 42), 55 copy-pasted chart files merged into 5 shared components; every step verified output-identical.
- **Teaching assistant**, company ChatGPT Team course (67 learners).

Shift Supervisor (值班股長) — Jan 2024 – Nov 2024

China Steel Corporation (中國鋼鐵) · Cold-Rolling Mill · Kaohsiung, Taiwan

- **Ran shift operations** on the cold-rolled coil line at **95% yield**.
- **Anomaly analysis** — SPC, trend and correlation analysis for root causes of defects and downtime; daily anomaly reports (Python / Excel).

— HOW EACH SYSTEM WAS VALIDATED

System	Method	Evidence
4 OCR systems	hand-labelled golden set; scored field-by-field in code (LLM only proposes)	**29 quotations · 32 / 31 invoices · 42 × 46 health fields**
n8n → skill migration	old vs. new output on the same inputs	**Excel 9/9 cells identical; PDFs byte-identical**
ESG refactor	API-by-API replay + pixel diff of charts after every step	**57 charts identical; dashboard never down**
Analysis platform (live)	engineers label each result; reviewed into the knowledge base	**feedback loop running on the BML line**

— RESEARCH & SELECTED PROJECTS

AIoT Pest Monitoring & Bee Pollination (M.S. thesis, NTU) — 2021 – 2023
Lead developer of greenhouse AIoT systems: bee tracking *99.5% mAP*, pest counting *95.8%* (YOLOv8, StrongSORT, Mask R-CNN).

Universal OCR Platform — no-code OCR builder (side project) — 2025 – 2026
Define fields and prompts, deploy as an OCR REST API in one click (FastAPI · React · Cloud Run); document analysis wrapped as MCP-format tools: extract document features → LLM routes to the best OCR template.